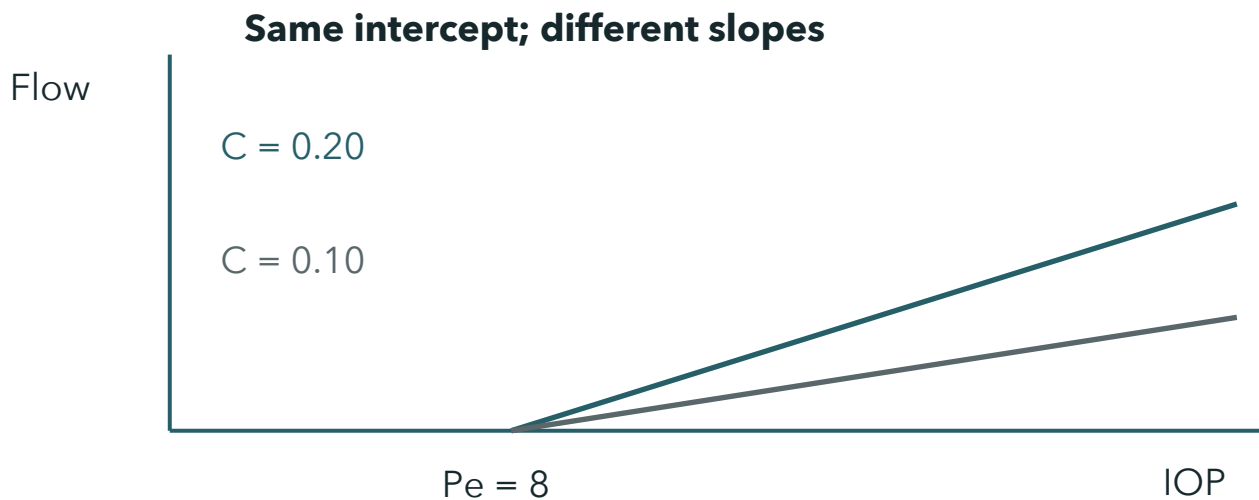


Aqueous dynamics and pressure gradients

Part 1 | A pressure-flow graph reveals the parameter

Slope and pressure-axis intercept encode different parts of conventional drainage.



Illustrative straight-line models share downstream pressure but differ in conventional outflow facility.

Read the axes

In the simplified relation $F = C(P - P_e)$, flow is vertical and pressure horizontal. C is the slope; P_e is the pressure-axis intercept. The model excludes pressure-insensitive drainage and assumes approximately steady conditions, so its fitted parameters should not be treated as universal constants.

Changing facility differs from changing P_e

Two interventions can raise IOP through different parts of the same equation.

At fixed flow

Feature	Change	Pressure consequence
Lower facility C	Larger F/C	Pressure rises
Higher P_e	Higher downstream baseline	Pressure rises
Higher secretion F	Larger flow demand	Pressure rises

Worked comparison

If conventional flow is 2 microlitres/min and facility is 0.25 microlitres/min/mmHg, the required pressure difference is 8 mmHg. Raising P_e from 8 to 12 mmHg raises model IOP from 16 to 20 mmHg without changing facility or secretion.

Add the route excluded by the first model

Pressure-insensitive outflow changes how secretion is partitioned.

Modified bookkeeping

- Total formation F
↓
- Subtract pressure-insensitive outflow U
↓
- Conventional component is $F-U$
↓
- $P=(F-U)/C+P_e$

Model boundaries

An assumed U can alter the inferred facility.

Real uveoscleral flow is not perfectly pressure-independent in every condition.

A fitted relation describes the measured range, not all possible pressures.

Pseudofacility is a measurement contribution

A pressure-decay estimate can include reduced inflow as well as true increased outflow.

An assumption fails

- Raise pressure during measurement
↓
- Independently observe reduced aqueous inflow
↓
- Pressure falls through net outflow minus inflow
↓
- Constant-inflow analysis attributes too much to outflow

Direction of the error

When the stipulated pressure rise suppresses inflow, the faster net fluid loss is not solely a change in outflow. Treating it as such overestimates true outflow facility. Measure or control inflow rather than assigning every pressure-dependent volume change to drainage.

Pressure is not the whole perfusion story

IOP participates in ocular perfusion, while vascular regulation also matters.

Comparison

Feature	Observation	Interpretive limit
IOP rises acutely	Perfusion pressure falls if arterial pressure fixed	Blood flow need not fall proportionately
Flow remains stable initially	Autoregulation may compensate	Reserve is finite
Different patients respond differently	Vascular state differs	One population average is insufficient

Postgraduate distinction

Separate the pressure gradient driving a vascular bed from the measured flow through it. A physiological compensation can preserve flow over a range; this does not establish that any arbitrarily high IOP is harmless or that all tissues autoregulate identically.